

SCE-004 — CALLING GRAPH EMERGENCE

Why Concepts Behave Like Expanding Calling Graphs Rather Than Isolated Labels

Structural Cognitive Emergence (SCE)

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with chatGPT (OpenAI)

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Repository: <https://github.com/sizhet/Structural-Cognitive-Emergence-SCE>

Introduction

Traditional educational systems often describe concepts as labels.

For example:

Cat

Dog

Bird

Tree

appear to be individual units of knowledge.

Similarly, many artificial intelligence systems treat concepts as categories, classes, or tokens.

However, everyday experience suggests something very different.

When a child learns:

Cat

the child does not merely learn a word.

The child immediately begins connecting:

Animal

Tail

Meow

Mouse

Tiger

Pet

Home

Sleep

Scratch

The concept rapidly expands into a network of relationships.

This observation motivates a central hypothesis of Structural Cognitive Emergence:

Concepts may behave more like expanding calling graphs than isolated labels.

The Label View

In the simplest interpretation:

Cat

is merely a category.

The brain learns:

Object

↓

Label

and stores the result.

This model is attractive because it is simple.

However, it fails to explain many aspects of human cognition.

For example:

Why does learning one concept immediately activate many others?

Why do concepts naturally generate associations?

Why do concepts lead to questions?

A simple label cannot explain these behaviors.

The Calling Graph View

Consider a different possibility.

Instead of:

Cat

being a single node,

suppose:

Cat

is actually a growing calling graph.

For example:

CAT

|

|—— Animal

|—— Tail

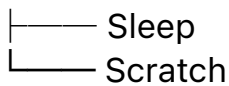
|—— Meow

|—— Pet

|—— Mouse

|—— Tiger

|—— Home



The concept is not isolated.

The concept is defined by its relationships.

Meaning emerges from connectivity.

Concepts Expand Naturally

One remarkable feature of childhood learning is that concepts rarely remain static.

A child first learns:

Cat

Soon afterward:

Tiger

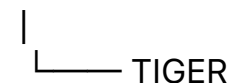
appears.

The child immediately notices:

Looks Like Cat

The graph expands:

CAT



Later:

Lion

appears.

Then:

Leopard

Then:

Animal

The structure continuously grows.

Concept formation therefore becomes graph growth.

Learning Is Graph Construction

This leads to a different interpretation of learning.

Traditional View:

Observation

↓

Classification

SCE View:

Observation

↓

Graph Expansion

Each observation contributes:

New Node

or

New Edge

The cognitive system continuously builds increasingly rich structures.

The objective is not merely recognition.

The objective is organization.

Differential Trees and Calling Graphs

The Differential Tree and Calling Graph perspectives complement each other.

Differential Trees answer:

What is similar?

What is different?

Calling Graphs answer:

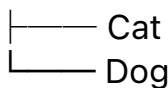
What is connected?

Together they form two views of the same cognitive landscape.

For example:

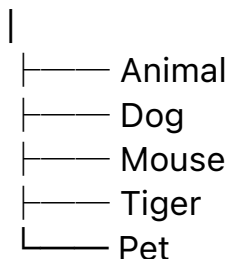
Differential Tree:

Animal



Calling Graph:

CAT



One organizes hierarchy.

The other organizes relationships.

Biological cognition may require both.

Concepts as Structural Regions

This perspective changes how concepts are defined.

Instead of:

Concept

=

Label

SCE proposes:

Concept

=

Calling Region

A concept becomes a local neighborhood inside a much larger cognitive structure.

The richer the neighborhood becomes,
the more meaningful the concept becomes.

Meaning is therefore relational.

Not isolated.

Why Children Learn Faster Than Labels Suggest

Children often appear to learn concepts extremely quickly.

This may occur because learning one concept automatically activates many neighboring concepts.

For example:

Cat

immediately activates:

Pet

which activates:

Home

which activates:

Family

which activates:

People

A single concept may therefore trigger growth throughout the network.

The graph expands in multiple directions simultaneously.

Learning becomes highly efficient.

The Emergence of Context

Calling Graphs naturally explain context.

Suppose a child hears:

Cat

inside a veterinary clinic.

The activated graph differs from hearing:

Cat

inside a toy store.

Different neighboring nodes become active.

Meaning therefore depends on the active region of the graph.

Context becomes a structural phenomenon.

Not merely a linguistic phenomenon.

Concepts Are Never Finished

An important consequence follows.

A concept is never complete.

Even an adult scientist possesses only a partial graph.

New experiences continue adding:

Nodes

Edges

Relations

The graph remains alive.

Knowledge becomes a continuously growing structure.

This observation aligns naturally with the idea of In-Situ Knowledge Growth.

Calling Graph Growth Creates Boundaries

As graphs expand,

they eventually encounter their limits.

For example:

CAT

becomes connected to:

Tiger

A new question appears:

Why are they similar?

Or:

Bird

becomes connected to:

Bat

A new question appears:

Why do both fly?

The graph has reached a boundary.

Something is missing.

A structural tension appears.

The Birthplace of Questions

This observation is critical.

Questions may not originate from ignorance.

Questions may originate from graph growth.

As structures expand:

New Connections

appear.

Simultaneously:

Missing Connections

become visible.

These missing connections create:

Structural Gaps

The appearance of a structural gap may be the true birthplace of:

WHY?

This idea becomes the central topic of the next document.

Implications for Artificial Intelligence

Most current AI systems excel at answer generation.

However, answer generation alone does not create cognitive structures.

Future AI systems may require explicit mechanisms for:

Graph Construction

Graph Expansion

Graph Maintenance

Gap Discovery

Only then may meaningful autonomous question generation become possible.

The Central Hypothesis

The central hypothesis of this document is:

Concepts are not isolated labels.

Concepts are expanding calling graphs.

In this view:

Observation

↓

Concept Formation

↓

Graph Growth

↓

Graph Expansion

↓

Boundary Formation

↓

Gap Discovery

Concepts therefore contain the seeds of future questions.

Every growing graph eventually encounters something it cannot yet explain.

And when that happens:

WHY?

appears.

Next

SCE-005

Gap Detection and Why Generation

How structural gaps may become the origin of curiosity, questions, and autonomous exploration.